

Supply chain uncertainty and diversification

THOMAS BOURANY
Columbia University/EIEF

IGNACIA CUEVAS
Analysis Group

GUSTAVO GONZÁLEZ
PUC Chile

Disclaimer: The views and opinions expressed are those of the authors and do not necessarily reflect those of the Central Bank of Chile or Analysis Group

Supply chain disruption risk and firms' sentiments

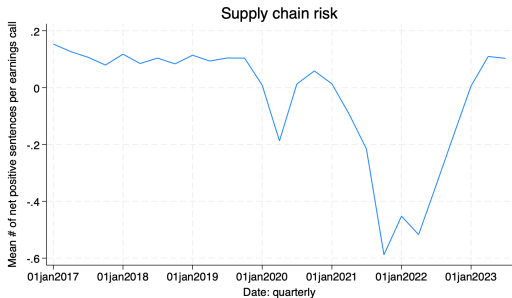


Figure 1: Average sentiment

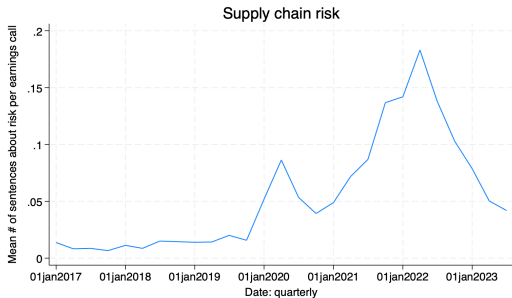


Figure 2: Average risk

- Hassan et al. (2023): Text-based measures on earning calls dedicated to the event of interest: *"Supply chain risk"*
 - Sentiment: perceived impact on the **mean** of the firm's economic outlook
 - Risk: perceived impact on the **variance** of the firm's economic outlook

Research question and results

- ▶ How does supply chain uncertainty affect firms' sourcing decision?
 - Are firms sourcing from foreign countries to diversify against risk? Or are they re-shoring?
 - Would firms import from cheaper or less risky countries?

Research question and results

- ▶ How does supply chain uncertainty affect firms' sourcing decision?
 - Are firms sourcing from foreign countries to diversify against risk? Or are they re-shoring?
 - Would firms import from cheaper or less risky countries?
- ▶ This project:
 - Multi-country sourcing model with supply chain idiosyncratic and aggregate *uncertainty*
 - Counterfactual analysis: changes in uncertainty and impact on firms sourcing, both on the *extensive* and *intensive* margin

Research question and results

- ▶ How does supply chain uncertainty affect firms' sourcing decision?
 - Are firms sourcing from foreign countries to diversify against risk? Or are they re-shoring?
 - Would firms import from cheaper or less risky countries?
- ▶ This project:
 - Multi-country sourcing model with supply chain idiosyncratic and aggregate *uncertainty*
 - Counterfactual analysis: changes in uncertainty and impact on firms sourcing, both on the *extensive* and *intensive* margin
- ▶ Preview of the results:
 - Theoretically, ambiguous effect of risk – four channels of expected profits:
 - ▷ *Option value* effect: risk raises the variance of sourcing capability, Oi-Hartman-Abel effect (+)
 - ▷ *Hedging* effect: covariance between sourcing capability & market demand (–)
 - Increase in uncertainty rises foreign sourcing: option value effect dominates the hedging effect, but aggregate ex-ante profits fall
 - Post-Covid-19: trade-cost *risk* (SD) rose – especially China and the world – driving entry into importing and a reallocation toward China

Related Literature

- ▶ Sourcing models
 - ▶ Antràs and Helpman (2004), Antràs and Helpman (2006), Antràs, Fort, and Tintelnot (2017), Bernard and Moxnes (2018)
 - Study supply chain disruption risk in multi-country sourcing model
- ▶ Uncertainty in trade
 - ▶ Theory: Grossman, Helpman, and L'Huillier (2023), Grossman, Helpman, and Sabal (2023), Gervais (2021, 2018)
 - Multi-country model allows for sourcing interdependencies and to separate effect of cost and uncertainty, aggregate and idiosyncratic.
 - Structural estimation with micro data and counterfactual analysis
 - ▶ Tariff policy uncertainty: Handley et al. (2020), Handley and Limão (2017), Charoenwong et al. (2023)
 - General framework for policy, supply-chain risk, and trade shocks
 - ▶ Trade disruption shocks Carreras-Valle (2021), Castro-Vincenzi (2022), Lafrogne-Joussier et al. (2022)
 - Analyze uncertainty and firm's sourcing choice using structural model
 - ▶ Empirical literature on propagation through trade networks: Caselli et al. (2020), Boehm et al. (2019), Carvalho et al. (2021), LaBelle et al. (2021), D'Aguanno et al. (2021)
 - Study of firms' joint sourcing and diversification decision

Stylized facts: supplier diversification after Covid-19

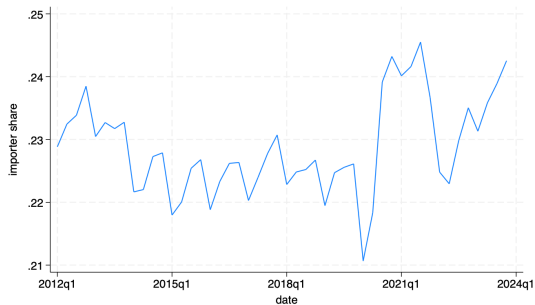


Figure 3: Share of importers among all firms

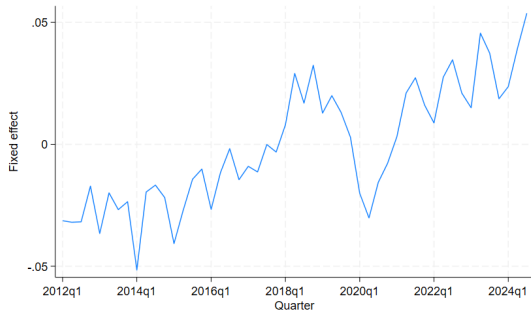


Figure 4: Number of sourcing origins per firm-product pair

- ▶ Chilean Customs $\times$ tax data, 2012–2023 (quarterly): ≈ 50 origins, $\approx 23\%$ of firms import
- ▶ Time fixed effects δ_t from $\log N_{fpt} = \alpha_{fp} + \delta_t + \epsilon_{fpt}$ (f firm, p HS6 product, t quarter)
Stable until 2017, rises for 2018–19 trade war, falls during Covid-19, then recovers *above* past levels

Stylized facts: entry into importing and substitution across origins



Figure 5: Share of importers from the US

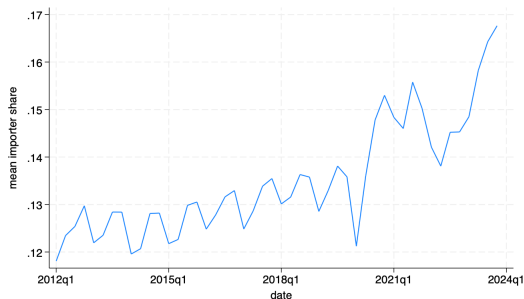


Figure 6: Share of importers from China

- ▶ Entry: share of importers trends down before 2020, then turns *up* – including among below-median-sales firms
- ▶ Substitution: the US share trends *down* while China's trends *up* – firms replace one origin for the other (avg. shares: US 0.098, China 0.128)

Model: Set-up and timeline

- ▶ Multi-country sourcing model inspired from [Antràs, Fort, and Tintelnot \(2017\)](#)
 - I countries. Origin: i , Destination: j .
 - Final-good producers in j : productivity φ , produce a single variety with monopolistic competition using intermediate inputs
 - Sourcing from set of countries $\mathcal{I}_j(\varphi)$
 - Trade from intermediate good firms in countries $i \in \mathcal{I}_j(\varphi)$, perfect competition, productivity follows Fréchet distribution with shape θ : $\mathbb{P}(a_i(\nu, \varphi) \geq a) = e^{-T_i a^\theta}$

Model: Set-up and timeline

- ▶ Multi-country sourcing model inspired from [Antràs, Fort, and Tintelnot \(2017\)](#)
 - I countries. Origin: i , Destination: j .
 - Final-good producers in j : productivity φ , produce a single variety with monopolistic competition using intermediate inputs
 - Sourcing from set of countries $\mathcal{I}_j(\varphi)$
 - Trade from intermediate good firms in countries $i \in \mathcal{I}_j(\varphi)$, perfect competition, productivity follows Fréchet distribution with shape θ : $\mathbb{P}(a_i(\nu, \varphi) \geq a) = e^{-T_i a^\theta}$
 - Supply chain risk from i to j (firm φ): [world-level](#), [country-level](#), and [firm](#) (idiosyncratic) shock

$$p_{ij}(\nu, \varphi) \equiv \tau_{ij} \underbrace{\tilde{\gamma}_j \hat{\gamma}_{ij} \tilde{\gamma}_{ij}(\varphi)}_{\equiv \gamma_{ij}} w_i$$

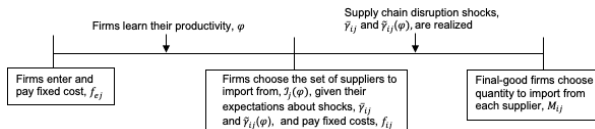
Model: Set-up and timeline

- ▶ Multi-country sourcing model inspired from [Antràs, Fort, and Tintelnot \(2017\)](#)
 - I countries. Origin: i , Destination: j .
 - Final-good producers in j : productivity φ , produce a single variety with monopolistic competition using intermediate inputs
 - Sourcing from set of countries $\mathcal{I}_j(\varphi)$
 - Trade from intermediate good firms in countries $i \in \mathcal{I}_j(\varphi)$, perfect competition, productivity follows Fréchet distribution with shape θ : $\mathbb{P}(a_i(\nu, \varphi) \geq a) = e^{-T_i a^\theta}$
 - Supply chain risk from i to j (firm φ): **world-level**, **country-level**, and **firm** (idiosyncratic) shock

$$p_{ij}(\nu, \varphi) \equiv \tau_{ij} \underbrace{\tilde{\gamma}_j \hat{\gamma}_{ij} \tilde{\gamma}_{ij}(\varphi)}_{\equiv \gamma_{ij}} w_i$$

- Household consumes final goods as a CES bundle with elasticity σ

- ▶ Three-stages firm decision:



Model: Firm Behavior Conditional on Sourcing Strategy, $\mathcal{I}_j(\varphi)$

- Share of intermediate input purchases for realized shocks:

$$x_{ij}(\varphi, \gamma) = \frac{T_i(\tau_{ij} \bar{\gamma}_j \hat{\gamma}_{ij} \tilde{\gamma}_{ij}(\varphi) w_i)^{-\theta}}{\Theta_j(\varphi, \gamma)} \text{ if } i \in \mathcal{I}_j$$

- ⇒ $\xi_i \equiv T_i(\tau_{ij} \bar{\gamma}_j \hat{\gamma}_{ij} \tilde{\gamma}_{ij}(\varphi) w_i)^{-\theta}$: *sourcing potential* of country i from the point of view of firm φ in country j
- ⇒ $\Theta_j(\varphi, \gamma) \equiv \sum_{k \in \mathcal{I}_j(\varphi)} T_k(\tau_{kj} \bar{\gamma}_j \hat{\gamma}_{kj} \tilde{\gamma}_{kj}(\varphi) w_k)^{-\theta}$: *sourcing capability* of firm φ in country j
- ⇒ *Ex-post* Eaton and Kortum, within the firm

Model: Choice of Set of Countries to Import from, $\mathcal{I}_j(\varphi)$

- Choice of $\mathcal{I}$ using ex-ante profits, with $\mathcal{I}_j(\varphi) = \{i : \mathbb{1}_{ij} = 1\}$:

$$\max_{\mathbb{1}_{ij} \in \{0,1\}_{i=1}^I} \mathbb{E}(\pi_j(\varphi, \gamma)) = \mathbb{E} \left(\underbrace{\varphi^{\sigma-1} \left(\eta \sum_{i=1}^I \mathbb{1}_{ij} T_i \left(\tau_{ij} \bar{\gamma}_j \hat{\gamma}_{ij} \tilde{\gamma}_{ij}(\varphi) w_i \right)^{-\theta} \right)^{\frac{\sigma-1}{\theta}}}_{\substack{\text{sourcing capability} \\ \Theta_j(\varphi, \gamma(\varphi))}} \underbrace{B_j(\bar{\gamma}_j, \hat{\gamma}_j)}_{\text{market demand}} \right) - w_j \sum_{i=1}^I \mathbb{1}_{ij} f_{ij}$$

- With market demand in j defined as

$$B_j(\bar{\gamma}_j, \hat{\gamma}_j) \equiv \frac{1}{\sigma} \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} E_j P_j(\bar{\gamma}_j, \hat{\gamma}_j)^{\sigma-1}$$

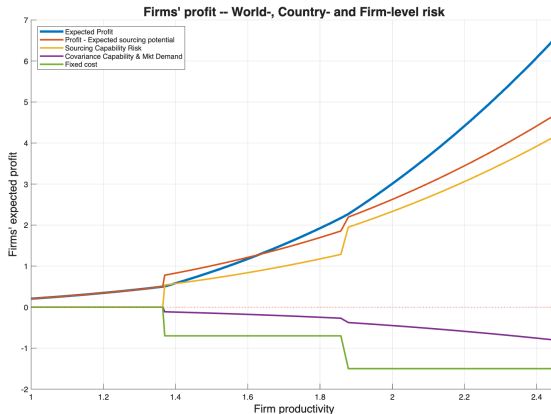
⇒ Effect through price index Price index

Decomposition: What determines profits?

- Expected profit decomposes into **four channels**:

$$\begin{aligned}
 \mathbb{E}[\pi_H(\varphi, \gamma)] = & \varphi^{\sigma-1} \left(\underbrace{\Theta_H(\varphi, \mathbb{E}[\gamma])^{\frac{\sigma-1}{\theta}}}_{\text{(i) Sourcing capability for expected cost}} \right. \\
 & + \underbrace{\mathbb{E}[\Theta_H(\varphi, \gamma)^{\frac{\sigma-1}{\theta}} - \Theta_H(\varphi, \mathbb{E}[\gamma])^{\frac{\sigma-1}{\theta}}]}_{\text{(ii) Option value (risk effect on capability)}} \Big) \times \underbrace{\mathbb{E}(B_H(\bar{\gamma}, \hat{\gamma}))}_{\text{(i-ii) Expected market demand}} \\
 & + \varphi^{\sigma-1} \underbrace{\text{Cov}(\Theta_H(\varphi, \gamma)^{\frac{\sigma-1}{\theta}}, B_H(\bar{\gamma}, \hat{\gamma}))}_{\text{(iii) Hedging effect (covariance)}} - \underbrace{w_H \sum_{i \in \mathcal{I}(\varphi)} f_{iH}}_{\text{(iv) Fixed cost of sourcing}}
 \end{aligned}$$

Profit Decomposition – 3 Countries Example



- ▶ Three countries, $\xi_0=1$, $\xi_1=0.8$, $\xi_2=0.3$, $(\sigma-1)/\theta = 1.58$.
World, origin and firm shocks Bernoulli: $\gamma=1$ w.p. 90%, $\gamma=5$ w.p. 10%
- ▶ Total (blue) = expected cost + option value + hedging – fixed cost, staggered by the pecking order.
Expected cost dominates: option value > hedging

Firms' sourcing strategies and the extensive margin

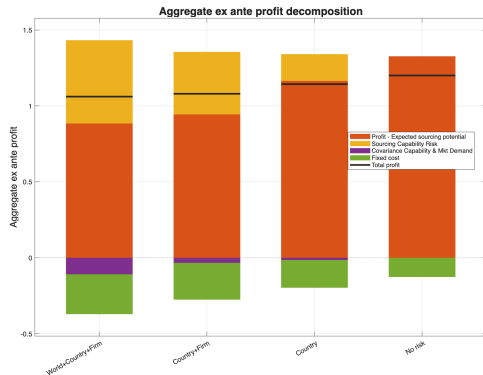


Figure 7: (a) Ex-ante profit decomposition

- ▶ Across risk scenarios (baseline = world + origin + firm risk $\rightarrow$ no risk), more risk $\Rightarrow$ more selection into importing: diversify to capture option values
- ▶ Aggregate ex-ante profit *falls* with risk: option-value gain $<$ expected cost + hedging effect + fixed costs

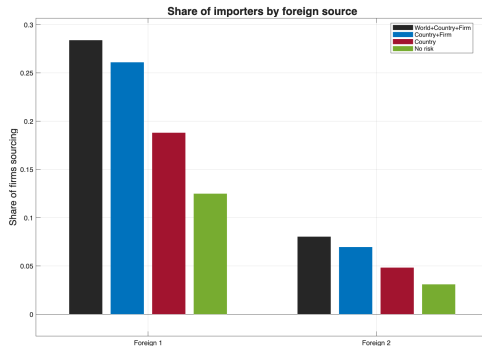
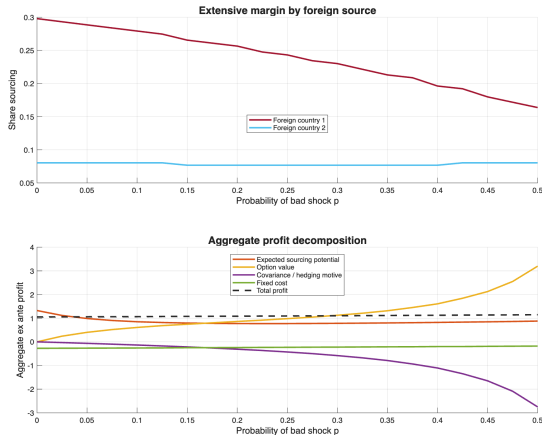


Figure 8: (b) Share of importers by source

Extensive margin and profit vs. risk



- ▶ Increase probability p of high-cost shock (raises both mean and variance of γ)
- ▶ Share importing from Foreign 1 falls in p (mean effect)
aggregate profit roughly flat: **expected cost** ↓ and **option value** ↑ offset by **hedging** ↓ and **fixed cost**

- ▶ Proprietary administrative data from the Central Bank of Chile:
 - **Customs** country-to-firm level data: import unit values, quantities, origin countries, and product categories
 - **Tax Form 29:** sales and materials purchases
 - **Unemployment Funds:** employer-employee level data.
 - Quarterly panel from 2012 to 2023.
- ▶ Publicly available data (yearly):
 - (i) CEPII gravity data set (population weighted distance, and common language): 2003-2021.
 - (ii) World Bank WGI (control of corruption): 2003-2021.

Structural Estimation: Step 1, Average Sourcing Potential

We estimate the parameters and shocks $\bar{\gamma}_{j,t}, \hat{\gamma}_{ij,t}, \tilde{\gamma}_{ij,t}^n, f_{ij}^n$.

- ▶ Take θ and σ from [Antràs et al. \(2017\)](#): $\theta = 1.789, \sigma = 3.85 \Rightarrow (\sigma - 1)/\theta = 1.583 > 1$
- ▶ Recall sourcing potential: $\xi_i^n \equiv T_i(\tau_{ij}\gamma_{ijt}^n w_i)^{-\theta}$
& import share: $\mathcal{X}_{ijt}^n = T_i(\tau_{ij}\gamma_{ijt}^n w_i)^{-\theta} / \Theta_{jt}^n$ if $i \in \mathcal{I}_j^n$
- ▶ Average component of sourcing potential: $\bar{\xi}_i \equiv T_i(\tau_{ij} w_i)^{-\theta}$
- ▶ Compare each firm's import share to its domestic share and estimate an origin fixed effect:

$$\log \mathcal{X}_{ij}^n - \log \mathcal{X}_{jj}^n = \alpha_i + u_{in}, \quad \alpha_i \mapsto \log \bar{\xi}_i$$

- ▶ Normalize $\xi_j = 1$ (no domestic disruptions). OLS on ≈ 50 origins using imports, wage bill, and input usage

Results

Structural Estimation: Step 2, Estimating Shocks

- ▶ Treat changes in relative import shares as changes in effective sourcing costs. The 4-quarter difference of log relative shares decomposes into three components:

$$\Delta_4 [\log \mathcal{X}_{ij,t}^n - \log \mathcal{X}_{jj,t}^n] = \underbrace{\bar{\lambda}_t}_{-\theta \Delta_4 \log \bar{\gamma}_{j,t}} + \underbrace{\hat{\lambda}_{it}}_{-\theta \Delta_4 \log \hat{\gamma}_{ij,t}} + \underbrace{e_{it}^n}_{-\theta \Delta_4 \log \tilde{\gamma}_{ij,t}^n}$$

- ▶ OLS with **time FE** (world $\bar{\gamma}_j$), **origin $\times$ time FE** (origin $\hat{\gamma}_{ij}$), and a **firm–origin residual** (idiosyncratic $\tilde{\gamma}_{ij}^n$)
- ▶ Recover levels assuming: log shocks follow a random walk. Initial value = 1 (no shock), log-normal $\Rightarrow$ mean and variance per component

Results, Chile-China/US

Structural Estimation: Step 3, Fixed Costs Estimation

- ▶ Use average 2012q1-2019q4 data: SMM for 13 countries (incl. Chile)
- ▶ Firm–origin fixed cost depends on gravity variables and control of corruption
 - $\log f_{ij}^n = \beta_c^f + \beta_d^f \log \text{dist}_{ij} + \beta_l^f \log \text{lang}_{ij} + \beta_C^f \text{corruption}_i + \nu_{ij}^n, \quad \nu_{ij}^n \sim \mathcal{N}(0, (\beta_{\text{disp}}^f)^2)$
 - We set $f_{jj}^n = 0$.
- ▶ SMM to estimate

$$\delta = [E, \beta_c^f, \beta_d^f, \beta_l^f, \beta_C^f, \beta_{\text{disp}}^f]$$

- ▶ Drawing shocks: Monte Carlo
 - Draw world, origin and firm shocks γ from log-Normal distributions (Quasi–Monte Carlo / Sobol sequences)
 - Draw firm productivity φ from a Pareto distribution.
 - Draw firm-level fixed costs f_{ij}^n from Normal distribution.

- ▶ Moment selection: m_k data, $\hat{m}_k(\delta)$ simulated
 1. Share of importers for all firms.
 2. Share of importers with firm sales below the median.
 3. Share of firms that import from each country.
 - ▶ $(I - 1) \times 1$ vector of moments.
 4. Share of firms whose input purchases from Chile are less than the median input purchases from Chile in the data.

- ▶ Model solution:
 - ▶ Equilibrium Price *Fixed point*,
 $\Rightarrow$ for each aggreg. shock (Monte Carlo) world $\bar{\gamma}_j$ + country $\hat{\gamma}_{ij}$, solve for $P(\bar{\gamma}, \hat{\gamma})$
 - ▶ Firm sourcing: high dimensionality *combinatorial* discrete choice problem
 $\Rightarrow$ For each firm φ , use Jia (2008)'s algorithm, in expectations (MC draws) to find $\mathcal{I}_j^n(\varphi)$

- ▶ Minimize $\hat{\delta} = \arg \min_{\delta} \hat{y}(\delta)^\top \mathbf{W} \hat{y}(\delta)$
 - ▶ We use $\hat{y}(\delta) = (\mathbf{m}/\hat{\mathbf{m}}(\delta) - 1)$ and assume $\mathbb{E}(\hat{y}(\delta_0)) = 0$ for the true parameter δ_0 .

Estimated supply-chain disruption shocks

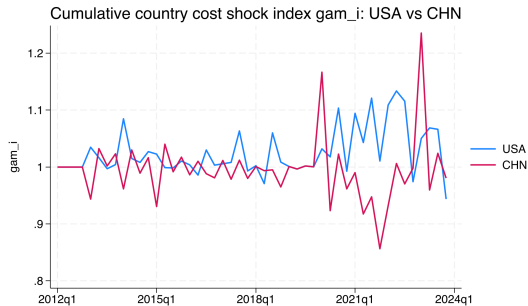


Figure 9: Source-specific shock $\hat{\gamma}_{ij,t}$: USA vs. China

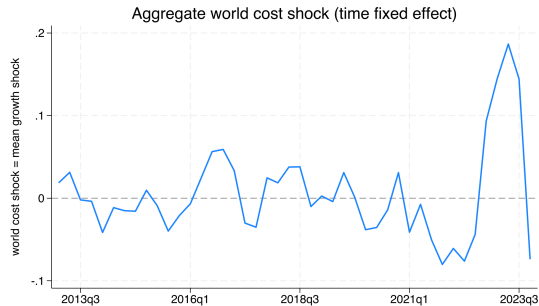


Figure 10: Global (world) shock $\bar{\gamma}_t$

- ▶ Higher $\hat{\gamma} \Rightarrow$ harder to source: post Covid, costlier to source from US and less from China
- ▶ World shock: trade became *easier* post pandemic (2021), then 2022 bottlenecks + Russia-Ukraine war raised costs

Aggregate shocks: mean and risk by period

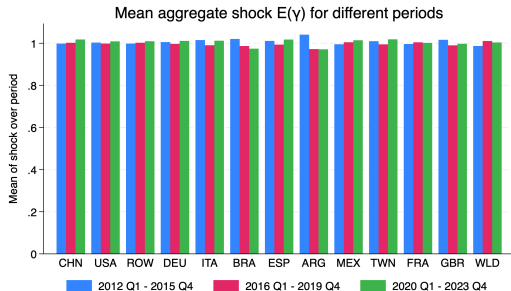


Figure 11: Mean $E[\bar{\gamma}_t]$, $E[\hat{\gamma}_{jt}]$

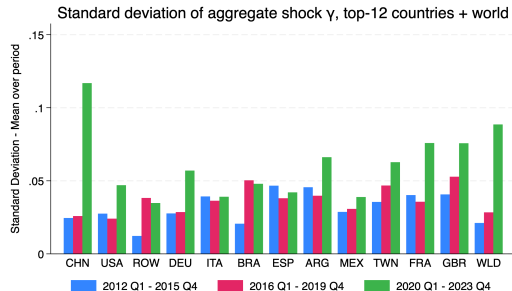


Figure 12: Std. dev. $\sigma(\bar{\gamma}_t)$, $\sigma(\hat{\gamma}_{jt})$

- Periods: 2012q1–15q4, 2016q1–19q4, 2020q1–23q4. Means rise modestly (China, US, world, trade-war peak in 2016–19)
- Post-Covid **risk (SD) rises markedly**, especially for China and the world – the key input to the counterfactual

Counterfactual exercise

- ▶ Estimate the model on 2012q1–2019q4 data: baseline sourcing and entry (**State 1**)
- ▶ Apply 2020q1–2023q4 trade-cost moments in three cumulative steps:
 - **State 2:** change the *means* $\mathbb{E}[\gamma_{ij}]$
 - **State 3:** add the change in *idiosyncratic* risk $\sigma(\tilde{\gamma}_{ij})$
 - **State 4:** add the change in *aggregate* risk $\sigma(\hat{\gamma}), \sigma(\bar{\gamma})$
- ▶ Evaluate effects on entry into importing and sourcing by country, and decompose aggregate ex-ante profits into the four channels

Post-Covid-19 uncertainty: firms' reactions

Table 1: Counterfactual states (2020–23 moments applied cumulatively)

State	Importer share	Origins/firm	China share	U.S. share
1: Baseline	0.1968	2.956	0.1430	0.0999
2: + Mean shock	0.1956	2.968	0.1404	0.0990
3: + Idiosyncratic risk	0.2095	2.832	0.1665	0.0958
4: + Aggregate risk	0.2135	2.815	0.1717	0.0963

- ▶ Risk (States 3–4) raises the **importer share**: more firms enter into importing
- ▶ Origins per firm falls: new importers source from a *single* foreign origin
- ▶ **China share rises**, U.S. share falls then partly recovers – reallocation toward China

Profit decomposition of post-Covid-19 uncertainty

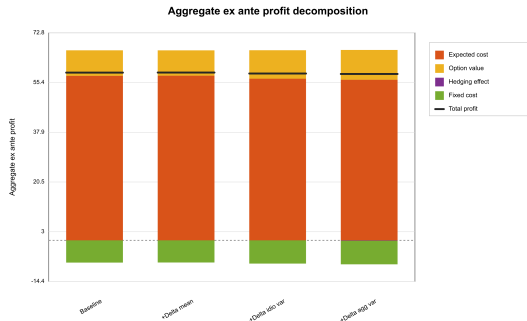


Figure 13: (a) Profit decomposition, 2020–23 uncertainty

- ▶ Expected-cost contribution *falls* (GE competition as more firms import). *Option value rises* and dominates
- ▶ Hedging effect & fixed costs become more negative $\Rightarrow$ aggregate ex-ante profit falls, as more firms diversify

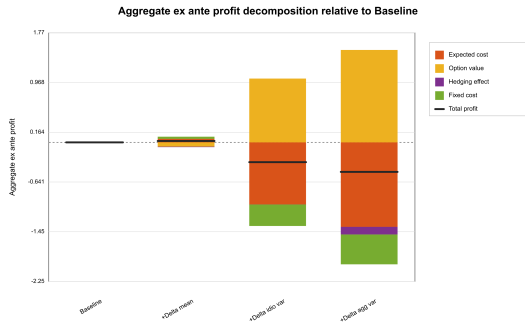


Figure 14: (b) Difference relative to baseline 2012–19

Conclusion

► Conclusion:

- a. Uncertainty shapes sourcing through **four channels**:
 - ▷ profits level is driven mainly by sourcing capability at expected costs
 - ▷ risk channels (option value, hedging effect) are an order of magnitude smaller
- b. Post-Covid-19, trade-cost *risk* rose sharply (esp. China and the world) while means moved less: firms responded by entering into importing and reallocating toward China
- c. The option value effect dominates the hedging effect, so firms diversify – yet aggregate ex-ante profits still *fall* (GE competition, hedging effect, and fixed costs)

► Next steps:

- Multi-sector model a la Caliendo Parro, with $\frac{\sigma-1}{\theta} < 1$ within sector
- Model with adjustment cost / export quantity frictions $\Rightarrow$ rationing
- Any feedback for the future of the project is very welcome!

Conclusion

► Conclusion:

- a. Uncertainty shapes sourcing through **four channels**:
 - ▷ profits level is driven mainly by sourcing capability at expected costs
 - ▷ risk channels (option value, hedging effect) are an order of magnitude smaller
- b. Post-Covid-19, trade-cost *risk* rose sharply (esp. China and the world) while means moved less: firms responded by entering into importing and reallocating toward China
- c. The option value effect dominates the hedging effect, so firms diversify – yet aggregate ex-ante profits still *fall* (GE competition, hedging effect, and fixed costs)

► Next steps:

- Multi-sector model a la Caliendo Parro, with $\frac{\sigma-1}{\theta} < 1$ within sector
- Model with adjustment cost / export quantity frictions $\Rightarrow$ rationing
- Any feedback for the future of the project is very welcome!
- Thank you! ***tb3219@columbia.edu***

References I

- Antràs, P., T. C. Fort, and F. Tintelnot (2017). The margins of global sourcing: Theory and evidence from us firms. *American Economic Review* 107(9), 2514–64.
- Antràs, P. and E. Helpman (2004). Global sourcing. *Journal of Political Economy* 112(3), 552–580.
- Antràs, P. and E. Helpman (2006). Contractual frictions and global sourcing. *NBER Working Paper* (12747).
- Bernard, A. B. and A. Moxnes (2018). Networks and trade. *Annual Review of Economics* (10), 65–85.
- Boehm, C. E., A. Flaaen, and N. Pandalai-Nayar (2019). Input linkages and the transmission of shocks: Firm-level evidence from the 2011 tōhoku earthquake. *Review of Economics and Statistics* 101(1), 60–75.
- Carreras-Valle, M. J. (2021). Increasing inventories: The role of delivery times.
- Carvalho, V. M., M. Nirei, Y. U. Saito, and A. Tahbaz-Salehi (2021). Supply chain disruptions: Evidence from the great east japan earthquake. *The Quarterly Journal of Economics* 136(2), 1255–1321.
- Caselli, F., M. Koren, M. Lisicky, and S. Tenreyro (2020). Diversification through trade. *The Quarterly Journal of Economics* 135(1), 449–502.
- Castro-Vincenzi, J. (2022). Climate hazards and resilience in the global car industry.

References II

- Charoenwong, B., M. Han, and J. Wu (2023). Trade and foreign economic policy uncertainty in supply chain networks: Who comes home? *Manufacturing & Service Operations Management* 25(1), 126–147.
- D'Aguanno, L., O. Davies, A. Dogan, R. Freeman, S. Lloyd, D. Reinhardt, R. Sajedi, and R. Zymek (2021). Global value chains, volatility and safe openness: Is trade a double-edged sword?. *Bank of england financial stability paper* 46.
- Gervais, A. (2018). Uncertainty, risk aversion and international trade. *Journal of International Economics* 115, 145–158.
- Gervais, A. (2021). Global sourcing under uncertainty. *Canadian Journal of Economics/Revue Canadienne d'Économie* 54(3), 1103–1135.
- Grossman, G. M., E. Helpman, and H. L'Huillier (2023). Supply chain resilience: Should policy promote international diversification or reshoring? *Journal of Political Economy* 131(12), 3462–96.
- Grossman, G. M., E. Helpman, and A. Sabal (2023). Resilience in vertical supply chains. *National Bureau of Economic Research*. (No. w31739).
- Handley, K. and N. Limão (2017). Policy uncertainty, trade, and welfare: Theory and evidence for china and the united states. *American Economic Review* 107(9), 2731–2783.
- Handley, K., N. Limão, R. D. Ludema, and Z. Yu (2020). Firm input choice under trade policy uncertainty. *National Bureau of Economic Research*. (w27910).

- Hassan, T. A., S. Hollander, L. Van Lent, M. Schwedeler, and A. Tahoun (2023). Firm-level exposure to epidemic diseases: Covid-19, sars, and h1n1. *The Review of Financial Studies* 36(12), 4919–4964.
- LaBelle, J., F. Leibovici, and A. M. Santacreu (2021). Global value chains and us economic activity during covid-19. *Available at SSRN 3885399*..
- Lafrogne-Joussier, R., J. Martin, and I. Mejean (2022). Supply shocks in supply chains: Evidence from the early lockdown in china. *IMF Economic Review*, 1–46.

- The price index is:

$$P_j(\bar{\gamma}) = \left(\int_{\tilde{\varphi}} \int_{\tilde{\gamma}(\varphi)} p_j(\varphi, \gamma)^{1-\sigma} d\tilde{\Psi}_j^{\varphi}(\tilde{\gamma}) dG_j(\varphi) \right)^{\frac{1}{1-\sigma}}$$

with:

$$p_j(\varphi, \gamma) = \frac{\sigma}{\sigma-1} \frac{1}{\varphi} \underbrace{\left(\eta \Theta_j(\varphi, \gamma) \right)^{-\frac{1}{\theta}}}_{mc_j(\varphi)}$$

$$\text{and } \eta = \left[\Gamma \left(\frac{\theta+1-\rho}{\theta} \right) \right]^{\frac{\theta}{1-\rho}}$$

Back

Proposition 1

1. Higher productivity firms will increase their expected profits by sourcing more from more or “better” countries:

$$\mathbb{E} \left(\Theta_j(\mathcal{I}_j(\varphi_H, \gamma(\varphi_H)))^{\frac{\sigma-1}{\theta}} B_j(\bar{\gamma}) \right) > \mathbb{E} \left(\Theta_j(\mathcal{I}_j(\varphi_L, \gamma(\varphi_L)))^{\frac{\sigma-1}{\theta}} B_j(\bar{\gamma}) \right)$$

2. If $\sigma - 1 > \theta$, import countries are **complements** in the sourcing decisions
 - ▶ More productive firms source from more countries
 - ▶ Pecking order: Same ordering of countries if fixed costs are the same across firms

Back

Closing the model: gravity and HHI

- ▶ Outside sector, freely tradable and big enough to pin down wages
- ▶ Using free entry condition and Fubini's theorem:

$$N_j = \frac{\alpha L_j}{\sigma \left(\int_{\tilde{\varphi}_j}^{\infty} \int_{\tilde{\gamma}(\varphi)} \sum_{i \in \mathcal{I}_j(\varphi)} f_{ij} d\Psi_{ij}^{\varphi}(\tilde{\gamma}) dG_i(\varphi) + f_{ej} \right)}$$

- ▶ Gravity:

$$M_{ij}(\bar{\gamma}) = \frac{E_j}{P_j(\bar{\gamma})^{1-\sigma} / N_j} \frac{Q_i}{\sum_k \frac{E_k}{P_k(\bar{\gamma})^{1-\sigma} / N_k} (\tau_{ik} \bar{\gamma}_{ik})^{-\theta} \Lambda_{ik}(\bar{\gamma})} (\tau_{ij} \bar{\gamma}_{ij})^{-\theta} \Lambda_{ij}(\bar{\gamma})$$

with $Q_i = \sum_k M_{ik}$ total production of intermediate inputs in i

- ▶ Model-implied HHI:

$$HHI_j = \sum_{i=1}^I \left(\frac{T_i (\tau_{ij} \bar{\gamma}_{ij} w_i)^{-\theta} \Lambda_{ij}(\bar{\gamma})}{\sum_{k=1}^I T_k (\tau_{kj} \bar{\gamma}_{kj} w_k)^{-\theta} \Lambda_{kj}(\bar{\gamma})} \right)^2$$

Parameterization

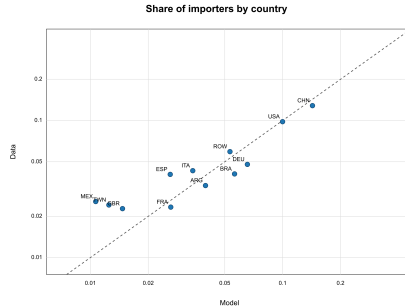
Variable	Definition	Value
γ shock	Bernoulli: $\gamma=1$ ($p=0.9$), $\gamma=5$ ($p=0.1$)	$\mathbb{E}[\gamma]=1.3$
ρ	Substitutability across intermediate varieties	2.00
I	Number of countries	3.00
ξ_D	Domestic sourcing potential	1.00
ξ_{F1}	Sourcing potential Foreign 1 (China)	0.80
ξ_{F2}	Sourcing potential Foreign 2 (US)	0.30
f_D	Fixed cost of sourcing Domestic	0.00
f_{F1}	Fixed cost of sourcing Foreign 1 (China)	0.70
f_{F2}	Fixed cost of sourcing Foreign 2 (US)	0.80
<i>Calibration for high complementarity $(\sigma - 1)/\theta = 1.58$ following Antràs et al. (2017)</i>		
σ	Elasticity of final demand	3.85
θ	Productivity Fréchet distribution shape	1.79

Back

Table 2: Targeted Moments

Moments	Data	Model
Share of importers	0.226	0.1959
Share imp. w/sales below median	0.082	0.0848
Median input purchases	124.430	112.56

Figure 15: Model fit: share of importers by country



Jia's Algorithm

- ▶ Proposition: $\forall i$, define mapping $V_{ij}(\varphi, \mathcal{I})$ to be one whenever including country i in sourcing strategy $\mathcal{I}$ raises firm-level expected profits $\mathbb{E}(\pi_j(\varphi, \mathcal{I}))$, and a value of zero o/w. Then, when $(\sigma - 1)/\theta \geq 1$, $V_{ij}(\varphi, \mathcal{I}') \geq V_{ij}(\varphi, \mathcal{I})$ for $\mathcal{I} \subseteq \mathcal{I}'$

- ▶ Algorithm:
 1. Let $V_i^n(\mathcal{I}) = 1$ if mg. ben. of adding country i is positive and 0 o/w

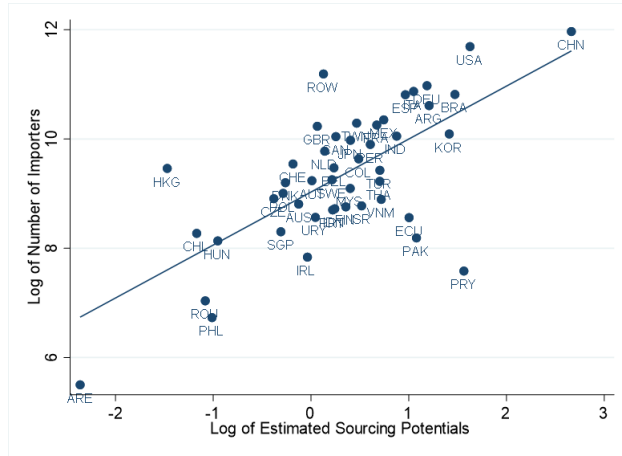
 2. When starting from set $\underline{\mathcal{I}}$ (no countries), iteratively add each country to the set (lower bound set)
 - ▶ Optimal sourcing strategy: minimum # of countries s.t. $V_i^n(\mathcal{I}) = 1$

 3. Then, starting from set $\bar{\mathcal{I}}$, remove 1-by-1 and find optimal sourcing str. (upper bound set)

 4. If sets do not overlap, only consider profits of upper bound set

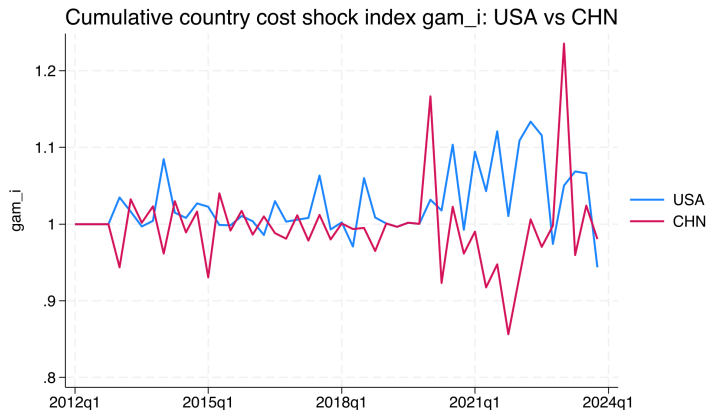
Estimated Sourcing Potential

Figure 16: Sourcing potential and extensive margin



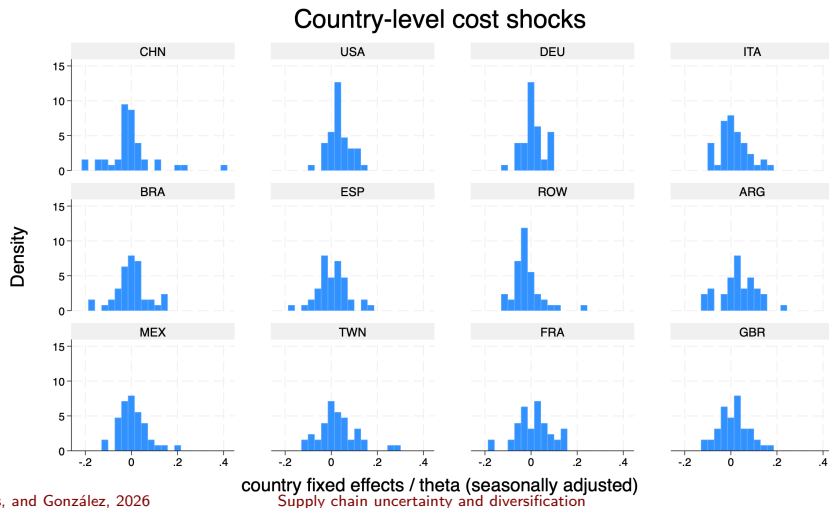
Estimated Source-Specific Supply Chain Disruption Shocks

Figure 17: Source-specific supply chain disruption shock $\hat{\gamma}_{ij,t}$: USA vs. China



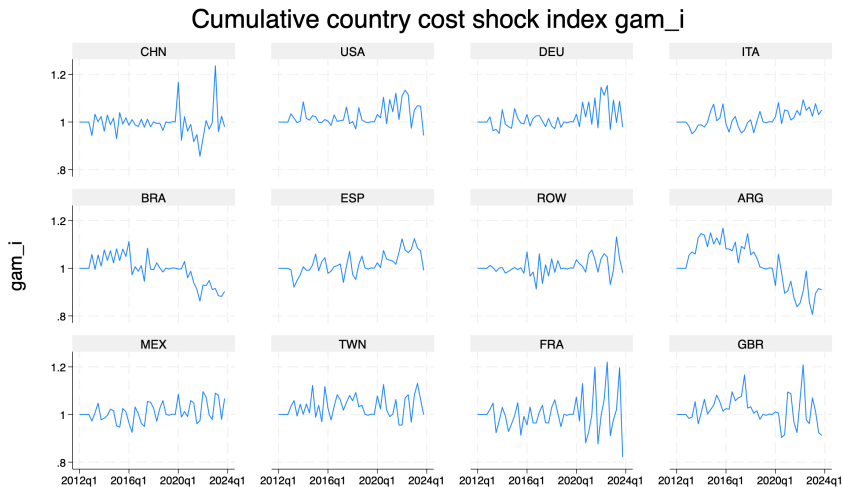
Estimated Source-Specific Supply Chain Disruption Shocks

Figure 18: Source-specific supply chain disruption shock $\Delta_4 \log \hat{\gamma}_{ij,t}$



Estimated Source-Specific Supply Chain Disruption Shocks

Figure 19: Source-specific supply chain disruption shock $\hat{\gamma}_{ij,t}$: all countries



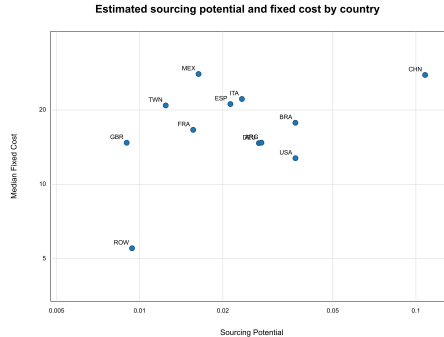
[Back](#)

Results: Fixed cost estimation

Table 3: Estimated Parameters (fixed costs in thousands USD)

E	f_{CHN}	f_{USA}	f_{ROW}	β_c^f	β_d^f	β_I^f	β_C^f	β_{disp}^f
256.6	27.72	12.76	5.51	1.73	0.277	1.122	-0.259	0.625

Figure 20: Estimated Sourcing Potential and Median Fixed Cost



Pecking order (data vs. random entry)

Table 4: Strings of countries followed by importers (top-10 origins)

String of countries	Data		Random Entry	
	Firms	% Imp.	Firms	% Imp.
CHN	7,970	23.68	1,865	5.54
CHN-USA	2,201	6.54	2,034	6.04
CHN-USA-RoW	348	1.03	664	1.97
⋮				
Total following pecking order	12,005	35.67	4,855	14.42

- ▶ 35.67% of top-10 importers follow a strict pecking order, vs. 14.42% under random entry
⇒ strong (but not perfect) ordering

Decomposition of average shares into margins

- ▶ Extensive margin:

$$\lambda_{ij} = \int_{\varphi} \mathbb{1}\{i \in \mathcal{I}_j(\varphi)\} dG(\varphi)$$

- ▶ Decomposition:

$$\bar{\chi}_{ij} \equiv \mathbb{E}_{\gamma} \left[\int_{\varphi} \chi_{ij}(\varphi, \gamma) dG_j(\varphi) \right] = \underbrace{\lambda_{ij}}_{\text{extensive margin}} \times \underbrace{\frac{\bar{\chi}_{ij}}{\lambda_{ij}}}_{\text{intensive margin}}$$

- ▶ We hence plot, for each country i , the decomposition:

$$\Delta\% \bar{\chi}_{ij} \approx \Delta\% \lambda_{ij} + \Delta\% \frac{\bar{\chi}_{ij}}{\lambda_{ij}}$$